

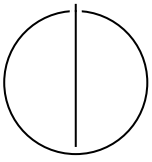
SCHOOL OF COMPUTATION,
INFORMATION AND TECHNOLOGY —
CHAIR OF HEALTH INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Bachelor's Thesis in Informatics

Decoding Music Perception: EEG-Based Genre Classification

Lukas Alex Jochim





SCHOOL OF COMPUTATION, INFORMATION AND TECHNOLOGY — CHAIR OF HEALTH INFORMATICS

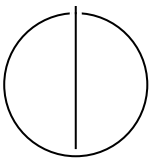
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Decoding Music Perception: EEG-Based Genre Classification

Dekodierung der Musikwahrnehmung: EEG-basierte Genre-Klassifizierung

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I confirm that this bachelor's thesis is my own work and I have documented all sources and material used.

Munich, 01.09.2025 Lukas Alex Jochim

Acknowledgments

Transparency in the use of AI tools

Abstract

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1 Introduction

1.1 Motivation

Maybe also mention here the gap between MGR and neurosciences more well, since it will be brought up in the Research goals section The human brain is one of the most complex and important systems within the human body. The brain manages various functionalities like emotion, memory, attention and sensory processing, to name a few. Apart from that, the brain plays a significant role in how we process musical pieces. Independent of culture and age, music is a universal tool to influence one's emotions, concentration and cognition. Furthermore, the brain gives us the unique ability to retrieve many different features out of music, like melody extraction or the identification of an artist or a specific genre. Depending on the style of music, the brain reacts differently. However, although the brain is central to cognitive processes such as listening to music, the core mechanisms remain unknown and are currently black boxes.

To enlighten these black boxes, medical advancements like functional Magnetic resonsance imaging (fMRI) or Electroencephalogram (EEG) arrived, which can measure brain activity. Especially EEG has aroused high interest since they directly measure the neural activity and can be done non-invasive with a little overhead in setup. This is why combining EEG and music has the ability to help better understand how our brain functions. By further adding means of Machine Learning (ML), a whole new interdisciplinary field opens up, bridging the gap between neurosciences and Computer science.

There is a growing research interest in the affective influence of music on the brain. However, classifying different music genres based on EEG signals remains underrepresented. This is exactly where this thesis aims to close the gap, by investigating whether it is possible to classify various musical genres depending on the EEG signals acquired using state-of-the-art ML techniques.

The research results have high potential in the domain of medical computer science. Especially with the growing investigations on Brain Machine Interfaces (BMIs), adaptive music therapy, personalised music recommendations or neurofeedback systems may become possible. Additionally, entertainment technologies can become more user-centred, facilitating unique experiences that impact many sectors like education and

mental health.

1.2 Research goals

1.3 Methodology

2 Theoretical Background

2.1 Medical foundations of EEG signals

2.2 Brain rythms

2.3 Correlation between music and brain activity

2.4 Recording and measurement of EEG signals

2.5 Denoising and preprocessing of EEG signals

2.6 Feature extraction and signal decomposition of EEG signals

3 Related work

3.1 Music Genre classification with audio features

3.2 EEG datasets with music stimulus

3.2.1 DEAP dataset

3.2.2 OpenMIIR initiative

3.2.3 MUSIN-G

3.3 Emotion classification with EEG signals

3.4 Music Genre classification with EEG signals

4 Music Datasets

Since this project deals with music genre classification, a suitable set of musical stimuli have to be chosen to gather valuable data. As research in the domain of music is widespread, there exist multiple common music datasets used in different studies. Therefore, the difficulty lies in choosing the right dataset for this thesis. However, the project goal cristallizes out clear requirements for such a music dataset.

First of all, the songs that will be listened to are suppossed to be around 30s long. The reason for that is that the stimulus could go lost if the song runs too long because the brain can adapt to the genre-specific features, thus reducing the visibility of the impact on the brain. Apart from that, the amount of songs should be quite large with around 800 to 1000 song clips. It is important to have a large pool of songs to get as many different and versatile data as possible. This helps in future steps regarding classification. However, related to this requirement, the genres used in this pool also have to meet specifications. First of all, the songs must be clearly labeled to avoid confusion in later stages of the study. This is also important to know for sure how balanced the dataset is in terms of genres listened to. This is supposed to be as balanced as possible so that each genre is represented equally. Additionally, it should be easy to employ this dataset in the study. It should be uncomplicated to load the songs as well as getting access to labels or other important information.

In the following, different popular and public music datasets are presented. They are going to be analyzed according their possible use in this research and on how they meet the requirements specified above. After that, the reasons for the final choice of dataset are outlined.

4.1 Audio Set

In 2017, Gemmeke et al. published a dataset for audio events called Audio Set[1]. This is a large-scale dataset consisting of audio events labeld by humans using a carefully structured hierarchical ontology of 632 audio classes. The source of the audio events are derived from 10 second segments of YouTube videos. These audio events are occurences in the real-world, like human sounds, animal sounds, natural sounds, sounds of things,

[1] `https://research.google.com/audioset/`

source-ambigious sounds, environment sounds and music. Then, each of these sounds is labeled with one ontology and labels with a maximal hierarchical depth of six. In the determination of labels, the researchers made sure that the tags are intuitive and clearly distinct from each other. [4]

In terms of the use for this project, the audio set has some drawbacks. First of all, the focus of this dataset does not lie in music but in general audio events. This is why in the first step, the dataset would have to be filtered to just contain the music clips. However, the fact that the clips are only 10 seconds long, there is the risk that this is not enough of a stimulus to arouse the brain and see the distinct patterns. Another problem lies in the format of the data. Since every audio event is a YouTube link, each sound would have to be downloaded from that link to allow a seamless integration with the experiment setup, thus illustrating a large overhead. Since the audio event labels range from many different occassions, only the audio events under the category of Music are important in this case. From this subset, only the songs are required that have one of seven genre labels, being Pop, Rock, Hip Hop, Techno, Rythm Blues, Vocal and Reggae. In total, there would be 40540 clips available. Unfortunately, the songs are not balanced out in terms of genres. [1]

4.2 Million Song Dataset

The Million Song Dataset[2] is a collection of audio features and metadata for a million music tracks, which also is freely available. At the moment, this is the largest research dataset in the field as it contains 1,000,000 songs accumulating around 280GB of data. In addition to the songs mainly from western commercial music in 2011, general metadata and audio features like pitch, timbre, etc. are provided by making use of the EchoNest API. This API is an online resource supporting many music applications on the web. Since most of the "natively" available metadata are coming from the EchoNest API, it is easily extendable to further labels like genres. [2]

Although this dataset may seem like a great fit because of its rich amount of songs, the amount of songs is too much. Since only 800-1000 songs are used during the acquisition, a majority of the dataset would have to be filtered out. Furthermore, all songs are not trimmed down to a shorter clip but are provided in their full length, which is not meeting the requirement for this study. Furthermore, the genres would have to be added externally since they are not available in the native dataset. This also emphasizes that it does not guarantee any balance in genres available over the course of the 1,000,000 songs.

[2] `http://millionsongdataset.com/`

4.3 MagnaTagATune

One challenge that comes up when creating such datasets is how a label or tag is determined for each song. In the context of genres, how is it decided which genre which song belongs to, as there could be overlapping features. On the basis of this hurdle, Law et al. created a music dataset to evaluate different tagging algorithms called MagnaTagATune[3]. It contains 25,863 music clips each 29 seconds long. These excerpts were taken from 5223 songs. These music pieces range across many different genres like Rock, Pop, Jazz, Blues, World, Metal, Punk and many more. In general, each clip is accompanied by a vector of binary annotations of 188 tags using the TagATune game. [5]

TagATune is a game with two players that aims to collect music tags from the players. Each player is presented with a music clip which they are assigned to give them labels. After that, both players see what the other player has chosen as tags. Based on this information, the players have to decide whether they listened to the same music clip. [6]

In perspective of this study, this music dataset has the advantage that each clip is around 30 seconds long, hitting the desired length for one stimulus. Furthermore, it is beneficial that the songs vary in genres used, thus being a good candidate as a music dataset. However, there are some drawbacks, especially lying in the labeling of each clip. Although the labeling process is well-thought and transparent, each song can have more than one label. This means that, depending on the players, one clip can be annotated with more than one genre tag. And since there are in total 37 genre related tags [7], deciding which genre is the most suitable choice for a song is very difficult. Related to that, since there are around 26,000 clips, one would have to remove majority of songs in order to let it fit into the scope of this thesis. Combined with the fact that multiple genres can be chosen for each song, it is difficult to achieve a balanced dataset according in regards of genre.

4.4 FMA Dataset

The Free Music Archive (FMA)[4] is an open dataset designed for multiple tasks in the domain of Music Information Retrieval (MIR). It consists of 106,574 tracks distributed over 16,341 artists and 14,854 albums. Although this dataset offers many different audio features and metadata about the songs, it is most tailored to music genres. Each track is arranged in a hierarchical taxonomy of 161 genres. As a labeling technique,

[3] `https://paperswithcode.com/dataset/magnatagatune`
[4] `https://github.com/mdeff/fma`

the researchers labeled the song according to the opinion of the artist who sings the respective song. Therefore, subjective errors in tagging and labeling could be minimized. In addition to that, all tracks are not only stored in simple .mp3 files, but also are fully licenced. To make the use of the FMA dataset easier, different subsets of the complete dataset are offered depending on the research question and scopes. In total, there are 4 subsets one can choose from. The first one is called *Full*, which has every track untrimmed and is therefore unbalanced in genre. The *Large* subset contains the same tracks as the *Full* set, but now, the audio is limited to 30 seconds each extracted from the middle of the track. To tackle the simpler problem of single-label prediction in genres, the *Medium* subset only contains those songs that only have one top genre, resulting in 25,000 30 second audio clips. Furthermore, the amount of genres is reduced from 161 to only 16. However, this subset still remains unbalanced. To solve this challenge, Defferrard et al. have constructed the balanced *Small* subset. It holds 8000 tracks each being 30 seconds long. From all the genres, the eight most popular ones were chosen and on the basis of this choice, 1000 clips were chosen for each genre. The eight genres in this case are Electronic, Experimental, Folk, Hip-Hop, Instrumental, International, Pop and Rock. [3]

The FMA dataset brings many favorable characteristics for this reseach project. The *Small* Subset of this set is a perfect fit since it is balanced according to genres and does not contain that many songs. However, around 900 songs per genre would need to be removed to have a reasonable amount of songs for the scope of this thesis. Another advantage is the recency of the publication. It is frequently used in different areas of MIR and is accompanied with many code examples and setup guides.

But, despite of being balanced in regards of the genres, the genres are not 100% clear resulting in unclear behaviour and characteristics of the songs. For instance, the genres International and Instrumental are very broad, since many genres can either have a cultural impact or don't have any lyrics. Another disadvantage is the fact the Classical music is not included, since this type of music is well-known for having an impact on the brain.

4.5 GTZAN Genre classification

Last but not least, there is the GTZAN Genre Dataset[5] which is a dataset specifically designed for Music Genre Recognition (MGR). There are 1000 songs in total originating from 2001 and 2002. Each track is trimmed to 30 seconds and range over 10 different genres. These genres are very common ones, including blues, classical, country, disco, hiphop, jazz, metal, pop, reggae and rock. Furthermore, each genre contains 100 songs,

[5] `https://www.kaggle.com/datasets/andradaolteanu/gtzan-dataset-music-genre-classification`

Dataset	Count of songs	average duration	Genres	Balanced option
Audio Set [4]	40540	10s	7	no
Million Song Dataset [2]	1000000	219s	undefined	no
MagnaTagATune [6]	25,863	29s	undefined	no
FMA Small [3]	8000	30s	8	yes
GTZAN [8]	1000	30s	10	yes

Table 4.1: Overview of potential Music datasets

resulting in a balanced song set in regards of genres. All audio was gathered from different sources like CDs, radio, microphone recordings and more to vary in the conditions of the recordings. Apart from the audio files, the dataset also contains audio features for each song. [8]

As this dataset is specialized on studies regarding MGR, it is a very good candidate for the thesis' purposes. With its 1000 songs, it fits perfectly in the right range of amount of songs. Moreover, since each genre consists of 100 songs, it is perfectly balanced out meeting another necessary requirement for the experiment. Another advantage is that the genres are clearly described from which clear attributes can be derived from. The only drawback in the choice of genres is that there are some genres, that are overlapping in characteristics. For instance, jazz originates from blues, thus sharing a lot of common characteristics. Nonetheless, many of the other genres have clear differences from one another, like classical and metal.

4.6 Final decision

All datasets presented in the previous sections have their strengths and drawbacks. An overview of the key attributes is shown in Table 4.1[6]. Based on the evaluation of each dataset for the project's purposes, it will be discussed which of the five datasets is most suitable as a source of songs for this thesis.

As already mentioned, the Audio Set does not fit well into the requirements of the project. Since its focus not even lies in music but in audio events, it is difficult to apply the set of clips in this use case. Moreover, ten seconds for each track is not enough for the purposes of arousing specific brain activity for each genre.

On the other hand of the spectrum, the Million Song Dataset is not beneficial either.

[6]The attributes only correspond to the parts of the dataset that is interesting in terms of MGR. The term undefined is used when there are no clear genre tags or if they depend on subjective interpretation and individual preprocessing

Although it only consists music tracks, they are not trimmed and are full length. Another disadvantage is the lack of balance in terms of genres. Apart from that, 90% of all songs would have to be removed since only around 1000 tracks are necessary for this study, which can be a great overhead (especially when trying to ensure a balanced set of songs).

From all the datasets, the MagnaTagATune dataset has the most sophisticated model of tagging each clip, which tends to higher quality in labels. Another benefit is that all songs are already trimmed to 29 seconds, meeting the duration requirement. However, this dataset is not focussed on MGR and consists of 188 music-related tags. Moreover, the tags are binary vectors, so that some songs could be labeled with more than one genres resulting in more difficult handling and unclear distinctions of songs and the genre they belong to the most. Out of this reason, the dataset is not chosen for the purposes of this research.

As the three previous datasets do not seem to be a good fit into the requirements, it has to be decided between the small FMA dataset and the GTZAN Genre classification dataset. Both datasets have very similar characteristics as they are specialized in MGR and therefore are balanced in genres, which is one of the most important requirements in this study. Additionally, all clips are availabe as audio files and have a constant duration of 30 seconds. The key difference lies in the size and genres present in the dataset. While the FMA set contains eight genres with 1000 songs each, the GTZAN contains ten genres with 100 songs each. Besides, in comparision to the genres represented in the GTZAN dataset, the genres in the FMA collection are not very descriptive and (for some of them) difficult to distinct from each other. For this very reason, the GTZAN song collection is chosen for the experiment setup and as a source of songs. Its easy file structure and naming convention can be seamlessly integrated into any setup, thus being the most suitable fit for this study.

5 Methodology

5.1 Acquisition

5.1.1 General setup, environment and stimuli

The experiment has been conducted in a laboratory setting. As mentioned in Chapter 4, the GTZAN Music dataset is chosen as a basis for the music stimuli. However, for the sake of the quality of the data, it would not be beneficial to hear all 1000 songs in one session. To avoid quick attention loss during a session and gather as much different data under different circumstances, the whole set of audio files is partitioned into 20 distinct sessions, each lasting around 30 minutes. This means that per session, 50 songs are played[1]. For each of these songs, not only are EEG signals measured, but also other biosignals like Photoplethysmogram (PPG), Inertial Measurement Unit (IMU) and temperature. However, this work will solely focus on the recorded EEG data. However, these 50 songs are not chosen randomly. Five songs from each genre are being played during the session to ensure that each of the ten genres from the dataset is represented in each session. This also ensures no genre bias in the data, thus resulting in a balanced signal across genres. During listening, the order of the songs is randomised to evoke a brain reaction that is as natural as possible, as the brain cannot predict the genre that is being listened to. Additionally, a five-second break is implemented in between songs to not overstimulate the brain.

All 20 sessions have been recorded over two weeks, with an average of two sessions per day. Between each session, there was a short break of between ten and 30 minutes. This verifies a sufficient pause of the brain to fully prepare for the next session. The test person is the author, a 21-year-old, right-handed college student without any common (brain) diseases. To further guarantee a high quality in the acquired signal data, the experiment's environment and setting must be adapted. First and foremost, the test person is instructed to sit comfortably in a chair with his eyes closed to minimise artificats coming from blinking and eye-movement. Furthermore and most importantly, any movements, especially in the head region, must be limited and as minimal as possible since the EEG device reacts very sensitive to head movements. Apart from that, the room was quiet and isolated with dimmed natural lighting combined with the

[1] Because of a corrupt audio file, one session only consists of 49 songs

brightness from a desktop monitor. The songs were played over the EEG headset (cf. Subsection 5.1.2) with average and constant volume.

5.1.2 Acquisition device

A non-traditional approach is chosen to record the EEG signal data coming from the brain. For this study, a prototype for a hybrid biosensing platform *Hugin* by Jabra is chosen. It is a Bluetooth headset that can play music and measure biosignals at the same time. For Audio, it uses classic Bluetooth to connect to a standard PC or Laptop. For the data, however, it uses a second low-energy bluetooth channel for the biosignal data, which can be connected to a second data acquisition PC.

The headphones have multiple sensors and electrodes to measure biosignals like EEG, PPG, IMU and temperature. For EEG, the headphones are supplied with 2 × 4 sensor electrodes and two bias electrodes. The sensors are easily configurable and provide an ultra-low-noise front-end, meaning that the data points are amplified with minimal distortion and interference. Besides that, the electrodes sample with 24-bit and a frequency of 250Hz.

For PPG, one optical sensor on the right side provides a high resolution of 19-bit ADC with a configurable sample rate. They also support raw data output or built-in algorithms to conclude heart rate or SpO2.

The IMU data is sensed by a 3-axis digital accelerometer fused with a 3-axis digital gyroscope. On a full scale, the accelerometer can recognize acceleration from $\pm 2g$ up to $\pm 16g$ (in steps of the power of two). The gyroscope can measure angular velocity from $\pm 125°/s$ up to $\pm 4000°/s$. The recognised data is then combined to provide accurate information on orientation, movements and position.

Last but not least, there are three digital thermometers, one on the skin, another one inside the earcup and one on the outside of the earcup to measure ambient temperature. All of those temperature sensors have an accuracy of ± 0.1°C.

Since this headset is still a prototype, they still is ongoing development and research on the headset. Nonetheless, some small-scale research projects already employed the biosignals from the headphones. These projects range from stress detection using the PPG data to device control and the recognition of cognitive fatigue using the EEG sensors.

5.1.3 Technical setup

Multiple technical components have to come together to gather the experiment's data. All the components employ the Lab Streaming Layer (LSL). It provides a standardised way to stream time-sensitive data between devices and applications. Devices or apps

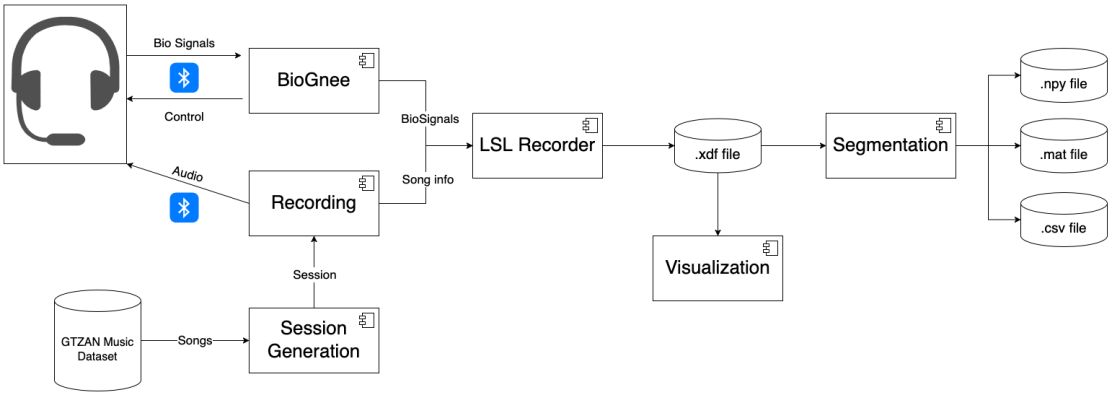


Figure 5.1: Technical setup

then push data on this stream, which is captured and stored in a special file format. The workflow is illustrated in Figure 5.1 and will be explained in the following.

At the core, there are two main programs responsible for the data. On the one hand, there is the Headset platform called *BioGNee*. This program is responsible for the sensor setup and starting and stopping the signal acquisition. Communicating with the headset's low-energy Bluetooth Channel reliably sets up the biosensors of the hybrid headphones, making the device ready for acquisition. After a successful setup, the data is pushed on the corresponding LSL Stream. On the other hand, the study uses a custom Python script. This Python script has two main tasks. Firstly, it loads all the songs from one of the 20 pre-generated sessions and randomly shuffles them. After that, the program plays each song one by one. Simultaneously, the script also pushes timestamps during recording to assign each segment to each song later.

The LSL Recorder combines both of these components. The LSL recorder collects all open streams on which the device/programs push their data. These are collected and stored in an XML-like file called .xdf. The resulting document, therefore, provides a unified way to not only store the signal data but also the respective time markers. These can be visualised by a custom MATLAB script to inspect the data quality. Important to note is that each distinct session occupies one .xdf file.

For future steps like Denoising or Feature Extraction, it is essential to look at each segment individually, i.e., each song. Therefore, a further custom Python program is used for the segmentation process of one session. Since the start and the end of one song are pushed to the LSL stream, one can read out the time these markers have been pushed and fetch all the biosignals data points measured in this interval. The resulting structure is a table with each song occupying one row. As columns, there is the genre on the one hand and the biosignals on the other hand. Each column related to biosignal

contains one channel from the respective Biosignal named after their defined name in the original .xdf file. The value of each column is then an array holding the collected data over the course of the song.

This table is then converted into three different file formats, .npy (numpy), .mat (MATLAB) and .csv. The numpy format is chosen for easy integration with the numpy library, since most of the data preprocessing is handled in python. To further support the MATLAB visualization function, the segments are stored in a .mat file to easily support the visualization of each segment, building on top of the existing Visualization subsystem. Last but not least, .csv is chosen since it is the state-of-the-art file format for reading, writing and processing data, which can become useful during Classification tasks.

5.2 Denoising

5.3 Feature extraction

5.4 Classification

5.4.1 Model 1

5.4.2 Model 2

5.4.3 Model 3

6 Results

6.1 Metric 1

6.2 Metric 2

6.3 Metric 3

7 Conclusion

7.1 Key findings and results

7.2 Problems and Limitations

7.3 Future work

Abbreviations

fMRI functional Magnetic resonsance imaging

ML Machine Learning

BMI Brain Machine Interface

FMA Free Music Archive

MIR Music Information Retrieval

EEG Electroencephalogram

PPG Photoplethysmogram

IMU Inertial Measurement Unit

LSL Lab Streaming Layer

MGR Music Genre Recognition

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